C# Arrays: Homework Basics

Tasks 1–3 practice array syntax with values given in code (no user input). Tasks 4–5 read input from the user and filter it.

Task 1: Array Basics

Create an `int` array with the values `3, 7, 1, 9, 4`. Print the first element, the last element, the element at index 2, and the length of the array.

Given in code	Print	Expected output
`3, 7, 1, 9, 4`	First element	`3`
	Last element	`4`
	Element at index 2	`1`
	Length of the array	`5`

Task 2: Sum and Average

Create an `int` array with the values `10, 20, 30, 40`. Use a loop to print all elements on one line, then calculate and print their sum and average.

Given in code	Print	Expected output
`10, 20, 30, 40`	All elements	`10 20 30 40`
	Sum	`100`
	Average	`25`

Task 3: Changing Values

Create an `int` array with the values `5, 10, 15, 20`. First change the element at index 1 to `100` and print the array. Then use a loop to multiply every element by 2 and print the array again.

Given in code	Step	Expected output
`5, 10, 15, 20`	After changing index 1 to 100	`5 100 15 20`
	After multiplying every element by 2	`10 200 30 40`

Task 4: Bigger Than 10

Ask the user how many numbers they want to enter, then create an array of that size. Read the numbers one per line and store them in the array. Then print only the numbers that are bigger than 10.

Input (size first, then one number per line)	Expected output
4 6 15 8 22	`15 22`
10 30 5 10 12 7 11 3 45 10 18	`30 12 11 45 18`

Task 5: Long Words

Ask the user how many words they want to enter, then create a `string` array of that size. Read the words one per line and store them in the array. Then print only the words that have more than 4 letters.

Input (size first, then one word per line)	Expected output
4 cat apple dog banana	`apple banana`
10 house tree river sky cloud mountain sea forest lake desert	`house river cloud mountain forest desert`
4 hi yes no ok	*(nothing is printed)*